GenAI in Medical Domain

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Certificate

This is to certify that the major project entitled **"GenAI in Medical Domain"** submitted by **Shubham Prajapati (Roll No: 24MCD017)**, towards the partial fulfillment of the requirements for the award of degree of Master of Technology in Computer Science and Engineering (Data Science) of Nirma University, Ahmedabad, is the record of work carried out by him under my supervision and guidance. In my opinion, the submitted work has reached a level required for being accepted for examination. The results embodied in this major project part-I, to the best of my knowledge, haven't been submitted to any other university or institution for award of any degree or diploma.

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Statement of Originality

I, **Shubham Prajapati**, Roll. No. **24MCD017**, give undertaking that the Major Project entitled "**Gen AI in medical Domain**" submitted by me, towards the partial fulfillment of the requirements for the degree of Master of Technology in **Computer Science & Engineering (Data Science)** of Institute of Technology, Nirma University, Ahmedabad, contains no material that has been awarded for any degree or diploma in any university or school in any territory to the best of my knowledge. It is the original work carried out by me and I give assurance that no attempt of plagiarism has been made.It contains no material that is previously published or written, except where reference has been made. I understand that in the event of any similarity found subsequently with any published work or any dissertation work elsewhere; it will result in severe disciplinary action.

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Date: 30/4/26

Place: Nirma University, Ahmedabad

Endorsed by

Dr. Swati Jain

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Abstract

Large Language Models (LLMs) have shown significant promise in medical question answering; however, purely model-based approaches often suffer from hallucination and lack the ability to justify their responses with reliable evidence [1]. Retrieval-Augmented Generation (RAG) addresses this limitation by combining language models with external knowledge sources, enabling grounded and evidence-based responses.

This work investigates the effectiveness of RAG-based systems in the medical domain and examines whether improvements in retrieval and reasoning can enhance both accuracy and reliability. Initially, a baseline system using a lightweight model (Flan-T5-Small) was evaluated with and without retrieval over a subset of the PubMed Central Open Access (PMC OA) corpus [2, 3]. Evaluation was conducted using the MIRAGE benchmark, which includes MedQA, MedMCQA, PubMedQA, BioASQ, and MMLU-Med [4, 5]. Building on this, the study extends to larger and domain-specific models such as BioMistral-7B, MedGemma-4B-IT, Typhoon-SI-Med-Thin, and II-Medica. A structured evaluation is performed across three configurations: Normal RAG, Advanced RAG with improved retrieval, and the proposed Agentic Corrective RAG (CRAG) framework. A controlled comparison using a consistent model is conducted to isolate the impact of different RAG architectures.

The CRAG framework introduces an intermediate validation step that classifies retrieved context into agreement, contradiction, and irrelevance before answer generation. This is supported by a hybrid retrieval strategy combining dense retrieval with BM25 and few-shot calibration to improve decision stability. Experimental results demonstrate that while retrieval-based improvements can enhance performance in certain datasets, they often introduce instability due to noisy or partially relevant context. In contrast, CRAG achieves more consistent and robust performance across diverse datasets by filtering unreliable information prior to generation. Overall, this work demonstrates that improving retrieval alone is insufficient for reliable medical question answering. Instead, integrating retrieval with reasoning and validation leads to both improved accuracy and enhanced stability, which are essential for deploying AI systems in safety-critical domains such as healthcare. This highlights the importance of combining retrieval, reasoning, and validation for building trustworthy medical AI systems.

Abbreviations

Abbreviation	Full Form
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
CRAG	Corrective Retrieval-Augmented Generation
PMC OA	PubMed Central Open Access
FAISS	Facebook AI Similarity Search
BM25	Best Matching 25 (Ranking Function)
QA	Question Answering
MMLU	Massive Multitask Language Understanding
MedQA	Medical Question Answering Dataset
MedMCQA	Medical Multiple Choice Question Answering
BioASQ	Biomedical Semantic Question Answering
PubMedQA	PubMed Question Answering Dataset
IT	Instruction-Tuned
JSON	JavaScript Object Notation
API	Application Programming Interface
GPU	Graphics Processing Unit
CPU	Central Processing Unit

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Chapter 1

Introduction

1.1 Background

The recent innovations in artificial intelligence, specifically Large Language Models, have made a significant contribution to the growth of NLP, showcasing an excellent capacity to understand complex questions and generate natural language answers. Transformer-based models utilize massive pretraining techniques to learn patterns in language and context, thus allowing to successfully complete numerous tasks using only little or no task-specific fine-tuning at all.

In healthcare, LLMs have already shown great promise for solving a variety of tasks including clinical question answering, summarizing biomedical literature, and supporting clinical decision-making processes.

At the same time, despite their remarkable skills, the usage of LLMs in medicine presents a number of serious challenges. The first one is the phenomenon known as "hallucination" whereby a machine produces seemingly logical but totally baseless responses. In a safety-critical domain such as healthcare, this problem may become especially acute, resulting in fatal mistakes. Another potential issue is associated with the inability of LLMs to transparently justify their actions and prove that provided solutions make sense.

In order to overcome these shortcomings, RAG models have been proposed as a potential solution. RAG models extend LLMs by making use of an external source of knowledge, which allows for the use of retrieved documents as the context to generate answers to the queries. While such an approach makes use of an effective technique to achieve factual grounding while reducing dependence on parametric memory, the success of RAG models largely hinges on the relevancy and quality of the retrieved data.

The results of the initial experiments carried out using a lightweight model and simple RAG pipeline indicated that retrieval does not necessarily result in an improvement in the model's performance. On the contrary, in a lot of instances, retrieval of irrelevant context was found to hinder the model's ability to make accurate predictions.

1.2 Motivation

However, the major issue associated with the proposed approach is that despite allowing making use of the context for accuracy and precision of factual information, several problems arise which make RAG less efficient. First and foremost, it is the problem of relevance of retrieved context which does not match the query posed. Furthermore, in some cases, relevant information is retrieved; however, it can be either ambiguous or contradict each other.

Due to the lack of the evaluation step associated with the retrieved context in traditional frameworks that assume the passive use of information provided to the system, the information could be used improperly by models resulting in wrong answers. In particular, it can be especially challenging for a model that processes medical queries which require precision.

Secondly, the second problem associated with context usage by models is related to their capacity to handle retrieved information. While the use of small models results in the problem of too much information, larger models would experience difficulties in distinguishing the relevance of context. Besides, the problem of inadequate formatting of the query compared to the format of retrieved information causes problems.

All these factors point to the need for an effective solution which will not only provide necessary information but also validate it. These motives become the foundation of CRAG system.

1.3 Problem Statement

However, despite the progress being made in both large language models and RAGs, there are several issues that have not yet found proper solutions. Namely, irrelevant and partial data and contradictions in the retrieved information could be included in the output of the information retrieval process as well; however, the issue of validating the context before generating the answer based on it remains without a solution.

Moreover, in the case of today's methods used in task evaluation, the main focus is placed only on the output itself, whereas no attention is paid to the reasoning process. In turn, it turns into an important issue when discussing medical applications of RAG-based systems.

Another major issue faced by modern RAGs involves the problem of unstable zero-shot reasoning. Specifically, during the process of reasoning, the model might display an unexpected behavior, such as considering the majority of the retrieved information contradictory to the others.

Therefore, one of the problems addressed by the current research paper consists in developing a system that is capable of answering medical questions using reasoning.

1.4 Objectives

One of the major goals of this project is to create a robust medical question answering system through the enhancement of both retrieval and reasoning modules. This will include exploring the effects of model size by testing small-domain and large-domain models. Additionally, a novel architecture for RAG will be designed that integrates context validation to mitigate noise in retrieval.

Additionally, enhancing the quality of retrieval through the combination of semantic and keyword search will be another aim of this research project. Other important goals are to identify and analyze failure modes in RAG systems such as classifier collapse and retrieval noise and offer solutions for addressing these problems.

Lastly, one goal of this project is to improve current evaluation methods by conducting trajectory analysis to provide insights into the answer generation process rather than just evaluating the accuracy of the outputs.

1.5 Research Questions

A number of research questions have been developed, and these are the forces behind the research process. One such question concerns conducting a research on how scaling affects the efficiency of the model in addressing medical queries via external knowledge.

Secondly, this will concern examining the relevance of retrieval for improving the efficiency of the model in terms of error generation during its implementation.

Thirdly, this will concern investigating whether there exists any possibility of verification of the information retrieved before using the information to answer the medical queries.

Lastly, it will concern finding out what else can be gotten from the examination of reasoning beyond the accuracy measure.

1.6 Scope and Contributions

The present research is a continuation of the previous project where multiple approaches were considered for increasing the reliability of the created medical Q&A system.

Among the important achievements of the study, one can mention the introduction of a new agentic corrective RAG (CRAG) pipeline. The paper provides the description of the CRAG pipeline and explains how the interaction between reasoning and retrieval contributes to the increase in the reliability of the findings. Besides, it carries out the controlled experiment to determine the impact of individual elements on the results' reliability and robustness.

Another major contribution achieved is related to the creation of a new hybrid approach to the retrieval of contextual information. In particular, this approach combines dense semantic search and BM25 keyword matching.

Moreover, in this paper, one of the key modes of failure, namely, 'classification collapse' is analyzed through the development of a technique that implements a few-shot calibration to enhance the stability of the model.

Moreover, the paper conducts a comparative analysis of Normal RAG, Advanced RAG, and CRAG variations to help analyze how different retrieval and reasoning mechanisms affect the performance of different sets of data.

Finally, a novel trajectory-based approach is proposed in this paper that enables us to analyze the decision-making process of the AI model.

In summary, the goal of this paper is to develop safe AI models for medical applications.

Chapter 2

Literature Review

2.1 Introduction

The LLMs have gained much attention for the application in the domain of medicine recently as well. The success in their abilities to comprehend complex questions in medicine and provide humane responses has been demonstrated. However, applying them directly in this sphere remains a problematic process since they may bring potential threats related to their accuracy, reliability, and safety issues.

This chapter represents an analysis of previous work on question answering systems in the domain of medicine, the method RAG, and new approaches for agent-based reasoning. This analysis will be helpful for identifying current problems and applying novel methods to develop a more reliable system.

Along with presenting previous approaches, this chapter will also be helpful for providing background information on experimental research conducted in this paper related to the improvement of RAGs.

2.2 Large Language Models in Medicine

Medical large language models (LLMs), including models based on GPT architecture, such as Med-PaLM [6], and domain-specific models, e.g., BioGPT [7], have achieved high accuracy rates when tested using medical benchmarks, namely MedQA and MMLU [8, 9]. They can perform well in answering complicated medical questions and even reach an expert level in some instances.

Nonetheless, there are still several drawbacks. First, these systems may make mistakes related to hallucinations, which means that they output facts that seem logical yet are untrue. In medicine, this problem may have fatal consequences. Secondly, LLMs tend to be opaque, meaning it is impossible to trace why an answer was given.

In such a case, the necessity of using models providing reliable evidence becomes obvious.

2.3 Retrieval-Augmented Generation (RAG)

RAG models were proposed as an attempt to overcome certain shortcomings of pure LLMs by incorporating external knowledge resources [10]. The model no longer uses only its internal knowledge and retrieves appropriate documents to provide more context for the generated answer.

This technology has become popular among many medical AI researchers. For example, in [11] and [12], authors show how RAG models can benefit from grounding in biomedical literature.

However, there are downsides to RAG technology as well. The output is highly dependent on the retrieved document. An incorrect document leads to a wrong answer regardless of other factors. Traditional RAG systems lack mechanisms for assessing the trustworthiness of the retrieved documents.

Moreover, recent experiments, such as the one conducted in the current paper, prove that RAG does not always help. In some cases, it even creates additional noise in the answer.

2.4 Benchmarking Medical RAG Systems

To facilitate benchmarking, several benchmarks like MIRAGE [4, 13] have been proposed. MIRAGE uses several corpora such as MedQA, PubMedQA, and BioASQ to establish a common framework for evaluating the RAG system.

The first key insight from MIRAGE is that performance gain cannot be guaranteed by applying RAG in all cases. Specifically, small models may fail to leverage retrieved context. This insight is aligned with the observation from previous sections of this project that lightweight models do not achieve significant gains from RAG.

This insight corresponds to some results presented in Chapter 4.

2.5 Retrieval Techniques

Retrieval plays a vital role in the working of RAG models and different approaches have been suggested to improve retrieval efficiency.

Dense retrieval relies on vector embeddings to ensure semantic similarities. Dense retrieval approaches work effectively in retrieving documents; however, they fail in cases where precise terminology is required.

On the other hand, the sparse retrieval model relying on techniques such as BM25 [14] relies on keyword-based approaches that perform effectively if medical terms are used precisely. Sparse retrieval approaches do not ensure semantic similarities.

However, both weaknesses are overcome using the hybrid approach that includes both dense and sparse retrieval approaches. In addition to both approaches, a re-ranking method is employed to rank results effectively using approaches such as ColBERT [15].

Such a hybrid method of retrieval has been gaining popularity recently, particularly in specialized areas including medicine. The methodology of the paper includes hybrid retrieval.

2.6 From Static to Agentic RAG

Traditional models for RAG adopt a static pipeline where the retrieved information serves as an input to the model. However, there have been numerous advancements in the literature towards more advanced and intelligent approaches, termed as agentic RAG.

In approaches such as ReAct [16], models are allowed to reason and perform actions such as determining whether more information is needed or what tool can be utilized. Therefore, instead of being confined to information retrieval, the model is able to determine whether or not that information is useful.

However, there have been some recent advances in the RAG pipeline where correction methods were integrated into the workflow. This reflects the fact that researchers seek to ensure that the RAG model remains reliable throughout the process by dealing with any retrieval errors before the generation stage. In light of this, the CRAG framework adopts similar techniques through context validation and filtering.

Corrective Retrieval-Augmented Generation (CRAG) First of all, Yan et al. [17], in their current research, develop a CRAG approach which provides assistance in ensuring the reliability of RAGs via evaluation of the retrieved documents prior to generating any response. Specifically, Yan et al. use a lightweight retrieval model to estimate the confidence score associated with retrieving certain information and, therefore, modify their process of retrieval, including changing a query itself or even applying an external search engine in some cases.

Additionally, CRAG leverages decompose-and-recompose strategies to extract meaningful information from documents obtained via retrieval. Such combination is supposed to address shortcomings of the classic RAG framework by addressing problems associated with erroneous retrievals.

However, the proposed CRAG approach mainly focuses on retrieving corrections, while, at the same time, ignoring the following reasoning stage. In contrast, the suggested methodology puts particular emphasis on reasoning through explicit classification of context into three categories, namely, agreement, disagreement, and irrelevance, and, therefore, facilitating precise filtering prior to generating answers.

The aforementioned approach becomes rather important when applied to scenarios associated with the healthcare domain due to potentially negative implications of misleading data on generated answers' accuracy.

2.7 Failure Modes in Medical AI

Reliability is an essential aspect in medical contexts. There have been some failure scenarios identified in the literature, which include hallucinations, over-confidence, and prompt design sensitivity.

One of the critical issues noted in this research includes the issue of classification collapse, where the machine wrongly classifies most of the data obtained as conflicting data.

This problem highlights the importance of calibration methods such as few-shot learning.

2.8 Recent Advances in Agentic and Self-Corrective Systems

Several studies have been conducted recently that look into reasoning beyond just once, and they explore the concept of iterative refinement and evaluation.

Self-Refine [18] presents a framework where the model can generate an output and refine it iteratively based on feedback from the model itself. It helps in improving the quality of answers generated; however, the process becomes computationally expensive.

Reflexion [19] builds upon this idea further by using memory and self-feedback to make the model aware of its previous mistakes and learn to improve itself while reasoning.

Toolformer [20] is another recent development that enables models to interact with external knowledge resources such as APIs and other third-party tools.

This reflects a new trend of developing agentic architectures that can reason, evaluate themselves, and refine themselves.

2.9 Validation and Verification in RAG Systems

Another emerging trend in recent studies involves the implementation of validation steps in the process of RAG.

Existing RAG models implicitly make an assumption about the reliability of the retrieved context. However, verification methods attempt to assess the retrieved data before generating answers based on them.

Some of them include:

- Filtering of context based on the relevance of retrieved items
- Verification of retrieved items using language models
- Reasoning pipelines for validating retrieved information

The proposed CRAG framework conforms to this trend by explicitly classifying retrieved context and discarding irrelevant data before answering the query.

2.10 Comparative Analysis of Existing Work

Table 2.1: Comparative Analysis of Medical QA, Retrieval, and Agentic RAG Systems

Work	**Model Type**	**Methodology**	**Key Strengths**	**Limitations**
Med-PaLM [6]	Large Medical LLM	Instruction-tuned transformer	Strong clinical reasoning capability	Requires large-scale computational resources
BioGPT [7]	Biomedical LLM	Domain-specific pretraining	Good biomedical knowledge representation	Limited reasoning and generalization
RAG [10]	Retriever + Generator	Dense retrieval with generation	Reduces hallucination via grounding	Highly dependent on retrieval quality
MEGA-RAG [11]	Medical RAG	Literature-grounded QA pipeline	Improved factual accuracy	High computational overhead
GuideGPT [12]	Clinical Chatbot	Context-aware retrieval-based QA	Domain-specific responses	Limited scalability and coverage
MIRAGE [4]	Benchmark Framework	Multi-dataset evaluation setup	Standardized medical QA evaluation	Resource-intensive evaluation process
FAISS [21]	Dense Retrieval	Vector similarity search	Fast and scalable retrieval	Struggles with exact keyword matching
BM25 [14]	Sparse Retrieval	Keyword-based ranking	High precision for exact terms	Lacks semantic understanding
ColBERT [15]	Neural Retrieval	Late interaction re-ranking	High retrieval precision	Computationally expensive
ReAct [16]	Agentic Framework	Reasoning + tool interaction	Enables dynamic decision-making	Sensitive to prompt design

Work	Model Type	Methodology	Key Strengths	Limitations
Self-Refine [18]	Iterative LLM	Self-feedback refinement loop	Improves response quality	Increased computational cost
Reflexion [19]	Agentic System	Memory + feedback-based reasoning	Enhances reasoning reliability	Complex implementation design
Toolformer [20]	Tool-Augmented LLM	API-integrated reasoning	Access to external knowledge sources	Training and integration complexity
CRAG [17]	RAG Enhancement	Retrieval evaluation + adaptive correction	Improves robustness to retrieval errors	Limited reasoning over retrieved context
CRAG (Proposed)	Agentic Medical RAG	Hybrid retrieval + validation + reasoning	Improved robustness, stability, and interpretability	Higher system complexity

This analysis demonstrates the evolution from classic RAG systems to sophisticated agentic systems. This indicates that although existing technologies enhance either retrieval or reasoning, the suggested technology combines both.

2.11 Research Gap

Despite some achievements, there still are some important areas that need further research:

- Underdevelopment of methods for incorporating retrieval correction and reasoning validation in medical RAGs
- Absence of precise mechanisms for evaluating context relevance and resolving contradictions
- Sole focus on evaluating the accuracy of results without assessing the reasoning process itself
- Inadequate attention to robustness and stability testing in varied medical datasets

Moreover, the majority of papers lack comparative studies of different RAG setups.

2.12 Position of This Work

This paper extends prior literature by developing an agentic approach to tackle the challenges mentioned above in the process of medical question answering.

Some of the main findings include the usage of hybrid search techniques, validation procedures, few-shot learning, and trajectory evaluation metrics.

In addition, this research also focuses on a comparative study involving Normal RAG, Advanced RAG, and CRAG variants.

Chapter 3

Methodology

3.1 Introduction

This chapter will explore the design and implementation process of the proposed question answering system for medical domain. The research will build on the standard architecture of Retrieval Augmented Generation (RAG) with better methods of retrieval and with inclusion of reasoning based techniques.

In order to systematically analyze the effect of the proposed improvements in RAG models, the model will be implemented in three different forms namely Normal RAG, Advanced RAG and CRAG (Agentic Corrective RAG).

3.2 Technical Background

Large Language Models (LLMs) utilize the transformer architecture that involves self-attention mechanisms to capture the dependencies within a sequence. This helps in generating outputs that are based on contextual understanding and relations.

In the medical domain, some applications include question answering, summarization, and reasoning. However, parametric knowledge of LLMs is likely to be inadequate and obsolete for some specialized domain queries.

To tackle this problem, the RAG framework integrates an external knowledge base. Instead of depending only on parameter-based knowledge of the neural network, documents from the external sources help the model generate accurate answers.

3.3 System Overview

The complete system comprises three major parts, which include retrieval, reasoning, and generation. The retrieval part collects pertinent documents, while the reasoning part determines the relevance of these documents, followed by the generation step that generates the final output.

Figure 3.1 provides a comparison between the conventional RAG architecture and the proposed CRAG architecture.

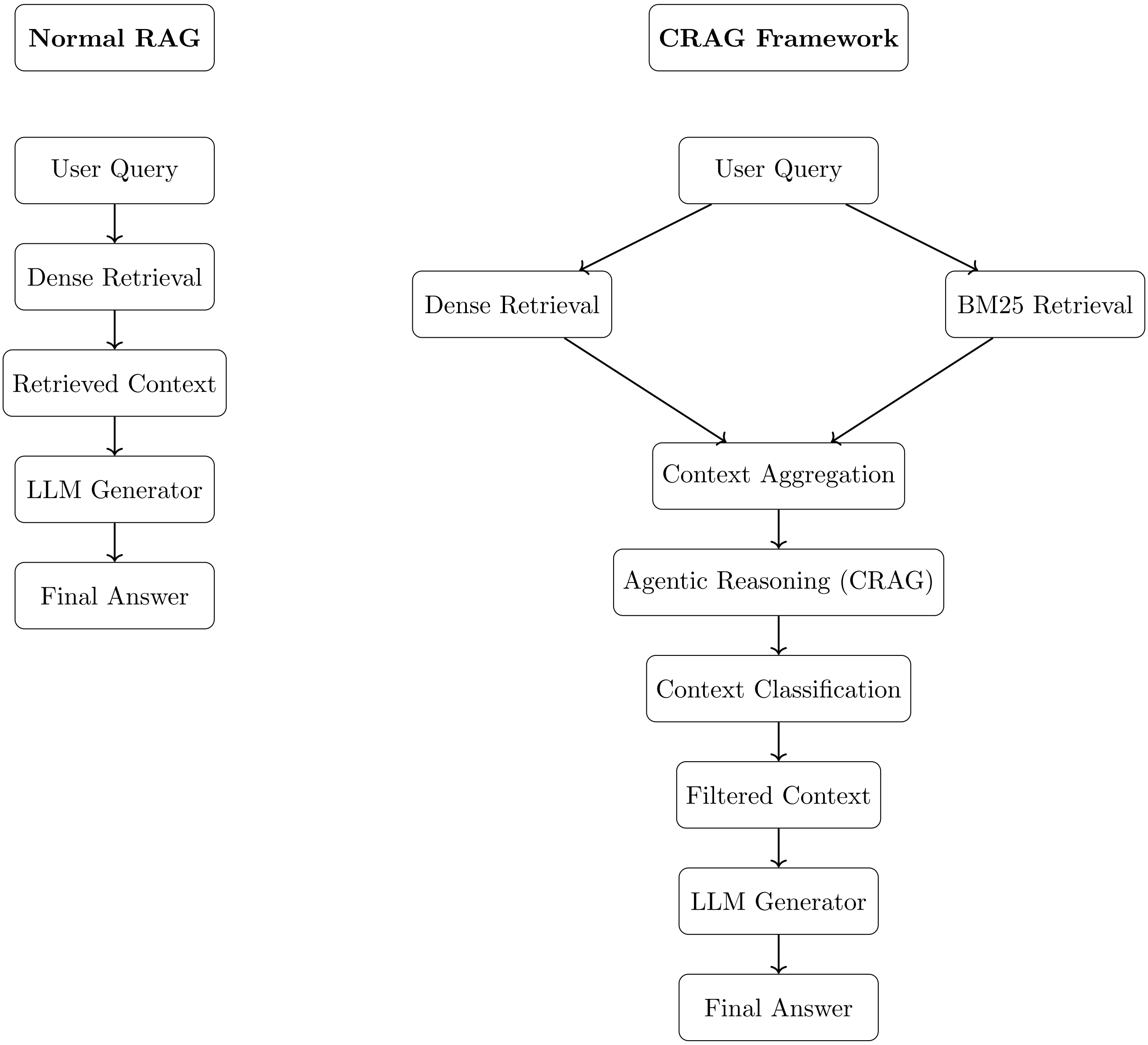

Figure 3.1: Comparison of Normal RAG and Proposed CRAG Architecture

3.4 CRAG Architecture Description

The agentic CRAG methodology enhances the conventional RAG architecture by adding a reasoning component between retrieval and generation. There are various components in the system that aid in making the system more robust and reliable.

3.4.1 Query Transformation

A user query is first transformed into a vector space using an embedding model. This aids in the process of semantic search over the knowledge base.

3.4.2 Dense Retrieval Component

The dense retrieval module conducts similarity search on the vectors to obtain the top-k relevant documents in the database. This helps capture the semantic relationship between the documents and the query.

3.4.3 BM25 Fallback Retrieval

In order to overcome the problem of dense retrieval in retrieving medical terminology, we propose a fallback retrieval approach based on BM25.

3.4.4 Context Aggregation

Contexts generated from dense retrieval and BM25 retrieval will be used to create a set of candidates for further analysis by the reasoning module.

3.4.5 Agentic Reasoning Module

This section is the most important part of CRAG. It involves evaluating retrieved contexts and assigning a label to each context. There can be three labels assigned to any context retrieved by the module, depending on its nature:

- **Agreement:** In case the retrieved context agrees with the query and is relevant to the task.
- **Contradiction:** When the retrieved context disagrees with the expected answer and is irrelevant.
- **Irrelevant:** The retrieved context is not helpful in finding the answer to the question.

These labels allow differentiating between useful and potentially misleading pieces of information.

3.4.6 Context Filtering

Irrelevant and contradictory passages will be filtered out after labeling. Only validated pieces of context will be used in answer generation. This allows avoiding the negative effect of retrieval noise.

3.4.7 Answer Generation Module

Filtered context will be sent to the language model responsible for generating the final answer. The validated context allows obtaining answers that are more reliable and accurate.

3.4.8 Logging of Trajectories

Every single step involved, such as retrieval outcomes, classification conclusions, and other output values, is logged in a systematic fashion. It is important to analyze the reasoning behind the decisions made by the system to make its results more interpretable.

3.4.9 Key Advantage of CRAG Framework

The key advantage of the CRAG framework lies in the possibility of separating the processes of retrieval and reasoning. Standard RAG frameworks pay attention solely to the process of retrieval; CRAG introduces another module for additional stability.

The elimination of unreliable context before generating answers ensures that the algorithm uses only valid information.

The standard process of using a Normal RAG framework does not include the validation step for the retrieved context before generating answers on its basis. The method, though simple, is highly vulnerable to any errors during the retrieval stage and may produce unreliable answers due to the invalid retrieved context.

On the contrary, CRAG involves a reasoning stage designed to exclude unreliable context before generating answers.

3.5 Agentic Reasoning Framework (CRAG)

CRAG architecture represents a development of RAG architecture which involves an additional reasoning stage in-between the process of searching and applying the searched context.

Unlike the traditional approach where the retrieved information is immediately used without any further checks, CRAG performs the analysis of the context retrieved on the basis of its belonging to one of the three types, such as agreement, contradiction, or irrelevance. It enables to eliminate false information prior to generating the context.

It should be noted that this approach was developed taking into account the fact that the retrieved context may often be contradictory, especially in the case of complicated queries like those in the field of medicine.

3.6 Workflow Details

The proposed system functions on the basis of a step-by-step procedure.

Firstly, the input query is analyzed and mapped to a vector representation. The vector representation is leveraged to fetch relevant passages from the vector database using dense retrieval.

To overcome the challenge of not being able to accurately capture exact medical terminology in semantic retrieval, we implement a fall-back system based on the BM25 model. This way, even key-word level matching is captured.

The retrieved passages are combined, and the list of passages is sent to the agentic reasoning step. Agentic reasoning analyzes the individual passages and classifies them as agreeing, contradicting, or irrelevant passages.

Depending on their classifications, unreliable passages are dropped. The remaining relevant passages are fed to the language model for generating the response.

3.6.1 Mathematical Formulation of CRAG

For the sake of clarity, let us define the proposed architecture in more precise terms, where we will denote the input query as q, and $R(q)$ as the set of documents retrieved using hybrid retrieval methods.

The filtering function employed under CRAG is $F(\cdot)$, whereby all the returned documents are sorted into one of the three categories: Agreement, Contradiction, or Irrelevance. Documents that have been categorized as Agreements are kept for consideration through the filtering process below:

$$F(R(q)) = \{d \in R(q) \mid d \text{ is an agreement}\}$$

This step is carried out to eliminate noise in the system. The final output of the model A is then determined by the language model taking into account the filtered retrieved documents:

$$A = \text{LLM}(q, F(R(q)))$$

From this formulation, it becomes obvious that the key difference between traditional RAG and the CRAG model lies in the fact that while RAG uses $R(q)$ directly, the CRAG approach involves the validation phase to retain only relevant content.

Implementation Mapping The classification phase based on the function $F(\cdot)$ is performed by the prompt-based classification using the language model, whereby each retrieved document gets assigned to the label agreement, contradiction, or irrelevance.

3.7 Datasets and Knowledge Base

The model is tested on several benchmarks like MedQA, MedMCQA, PubMedQA, BioASQ, and MMLU-Med. These datasets are of varied kinds of questions ranging from exam-type questions to literature type questions.

The knowledge base is generated using the PMC OA dataset. The documents in the dataset undergo several processes such as cleaning, segmenting, and generating embeddings.

3.8 Retrieval Mechanisms and Hybrid Strategy

The retrieval process involves a hybrid strategy involving dense and sparse retrieval techniques.

In dense retrieval, there is capturing of the semantic relatedness between the query and documents through vectors. In sparse retrieval, achieved using the BM25 technique, there is making sure that no exact keyword match is missed.

The scoring function for BM25 takes the form:

$$\text{score}(D, Q) = \sum_i IDF(q_i) \cdot \frac{f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1(1 - b + b \cdot \frac{|D|}{avgdl})}$$

The rationale behind the choice to use hybrid retrieval is the weakness in both retrieval techniques. Despite dense retrieval capturing the meaning, there can be overlooking of critical domain-related words, hence the need for BM25.

3.9 Experimental Configurations

For testing the system performance, the following configurations have been chosen:

Normal RAG – in this case, the context retrieval stage is followed by document processing which is used directly for answer generation without context validation.

Advanced RAG configuration aims at improving document retrieval stage through the use of improved embedding and context re-ranking methods. This allows for reducing the noise in the retrieved context.

Finally, in the CRAG configuration, we introduce an agentic reasoning component aimed at verifying the retrieved context before processing it further.

3.10 Models and Design Considerations

Models selection for the system evaluation is done according to criteria related to efficiency, performance, and domain specialization capabilities.

For the initial set of experiments, lightweight models (Flan-T5-Small) were chosen due to efficiency. More powerful models (BioMistral-7B, II-Medica) were chosen because of their ability to perform complex inference and incorporate specialized knowledge.

Also, instruction-tuned models (MedGemma-4B-IT) proved to be very good in understanding structured prompts and generating appropriate context-aware responses.

Different models allow us to test the effect of model scale and specialization.

3.11 Experimental Design Considerations

All experiments use the same datasets, evaluation criteria, and parameters for the retrieval procedure to allow a fair comparison between different configurations.

In all experiments, the number of documents retrieved (the top-k) remains the same across different configurations. Also, prompt formats are used consistently to reduce variability caused by prompt engineering.

Such an experimental design enables us to attribute performance differences solely to the changes in retrieval and reasoning.

3.12 Solving Problems of Classification

Initially, it was discovered that the model had difficulties distinguishing between agreement, contradiction, and unrelated documents since there was a high chance of classifying context as contradictory.

This problem is addressed by calibrating the model with a few training data points to make it more robust.

3.13 Implementation Details

Modularity was used in implementing the system to allow easy experimentation.

The knowledge base is built based on the content from the PMC OA articles, which have been transformed into embeddings. These embeddings are stored in a vector database to enable similarity search.

The reasoning engine uses prompt-based classification where intermediate results such as the retrievals and classifications are logged.

3.14 Evaluation Methodology

All models are evaluated using multiple choice questions where accuracy is the main metric.

$$Accuracy = \frac{\text{Number of Correct Answers}}{\text{Total Number of Questions}}$$

3.15 Trajectory Tracking

To have more insight into the behavior of the system, a trajectory tracking algorithm is introduced. This captures the various intermediate steps such as retrieval outputs, classifications, and triggerings of back-ups.

This allows for investigation of the reasoning process and provides information about how the system reaches its conclusions.

3.16 Computational Complexity and Efficiency

The computational complexity of the system lies in two areas, namely retrieval and generation.

In dense retrieval, there will be computation of vector similarities depending on the number of documents in the corpus. There are some efficient methods of indexing that can lower the computational burden.

The computational cost of BM25 retrieval is smaller, although this will necessitate the examination of more of the corpus.

Reasoning incurs extra computational costs, but this is worth it since reliability is increased.

3.17 Limitations of Methodology

The accuracy of the suggested approach relies heavily on the quality of document retrieval and prompt-based classification. Any issues at these stages can have an impact on the final output.

Another assumption made by this method is that all necessary facts can be found within the knowledge base, which may not necessarily be true in practical applications.

3.18 Pipeline Overview

The entire process involves query parsing, hybrid retrieval, context aggregation, filtering based on reasoning, and finally, answering.

Chapter 4

Results and Analysis

4.1 Experimental Results and Discussion

The current chapter entails a thorough evaluation of the experimentally achieved results for varying setups of the proposed medical question answering system. The experimental setup is designed in such a way that it follows a certain progression, starting from a basic setup of the system with a standard Retrieval-Augmented Generation pipeline, then followed by its advanced version called Advanced RAG and finally, the proposed Agentic Corrective RAG.

All experiments were carried out on several medical benchmark datasets, namely MedQA, MedMCQA, PubMedQA, BioASQ, and MMLU-Med. The datasets encompass structured and unstructured exam-style as well as literature-related questions to evaluate system effectiveness properly. The primary metric employed for evaluating system performance is accuracy.

Note on Experimental Setup While the initial results include evaluation across several models to study their behavior, further results employ a single model called II-Medica in order to conduct a controlled evaluation across RAG systems.

Experimental Setup The results are shown in two steps. At first, several models are evaluated employing a standard RAG setup. Then, a single model (II-Medica) is evaluated in the course of a controlled test across RAG setups.

4.2 Normal RAG Performance

The findings from the Normal RAG setting show considerable variation among different models and datasets. The variation shows the reliance of RAG systems on their model architectures and dataset qualities.

II-Medica exhibits the best balance of performance, thus being ideal for comparative purposes

Table 4.1: Normal RAG Performance Across Models

Model	MedQA	MedMCQA	PubMedQA	BioASQ	MMLU
BioMistral-7B	0.3291	0.3519	0.3640	0.6214	0.3673
Qwen1.5-7B	0.2773	0.3223	**0.5520**	**0.6392**	0.2158
Typhoon-SI-Med-Thin	**0.5656**	**0.5333**	0.4220	0.1294	**0.7181**
II-Medica	0.4438	0.4583	0.5440	0.6084	0.6465

in subsequent sections.

4.3 Advanced RAG Performance

Table 4.2: Advanced RAG Performance (II-Medica)

MedQA	MedMCQA	PubMedQA	BioASQ	MMLU
0.5010	0.4790	0.4120	0.5020	0.6750

In Advanced RAG, the effectiveness of the system is enhanced via better embedding and re-ranking methods. As a result, the effectiveness is increased in datasets that require more reasoning, like MedQA and MMLU.

But, effectiveness drops in literature-based datasets like PubMedQA and BioASQ, proving that better retrieval does not necessarily mean better answers.

4.4 CRAG Performance

Table 4.3: Agentic CRAG Performance (II-Medica)

MedQA	MedMCQA	PubMedQA	BioASQ	MMLU
0.5145	0.5044	0.4740	0.6262	0.6873

CRAG presents a reasoner component to assess the retrieved context before providing an answer. It results in consistent performance across different datasets.

4.5 Controlled Comparison of RAG Variants

The robustness of CRAG is evident through its consistent reduction of performance variance across different datasets, especially under noisy retrieval conditions.

Table 4.4: Performance Comparison Using Same Model (II-Medica Across RAG Variants)

Dataset	Normal RAG	Advanced RAG	CRAG
MedQA	0.4438	0.5010	**0.5145**
MedMCQA	0.4583	0.4790	**0.5044**
PubMedQA	**0.5440**	0.4120	0.4740
BioASQ	0.6084	0.5020	**0.6262**
MMLU	0.6465	0.6750	**0.6873**

Table 4.5: Percentage Improvement Across RAG Configurations (Same Model)

Dataset	Normal → Advanced	Advanced → CRAG	Normal → CRAG
MedQA	+12.9%	+2.7%	**+15.9%**
MedMCQA	+4.5%	+5.3%	**+10.1%**
PubMedQA	-24.3%	+15.0%	-12.9%
BioASQ	-17.5%	+24.7%	**+2.9%**
MMLU	+4.4%	+1.8%	**+6.3%**

4.6 Performance Improvement Analysis

General Observation The Normal → CRAG table reflects the overall improvement attained by the proposed solution. Despite Advanced RAG introducing instability for certain data sets, CRAG ensures that performance is restored and delivers more consistent results.

Analysis on Stability Along with increased accuracy, the CRAG system proves to be more stable, delivering more consistent results across different data sets. Unlike Advanced RAG, which experiences considerable instability by improving performance in one set and worsening it in another, CRAG offers more stability.

The reason behind such stability lies in the context validation process, which eliminates the negative influence of incorrect retrieval. Thus, CRAG ensures both higher accuracy and greater stability for questions of different natures.

4.7 Analysis of PubMedQA Behavior

Although PubMed-based documents have been used as the knowledge source, the performance of PubMedQA declines with Advanced RAG.

This is due to the fact that PubMedQA involves reasoning from scientific abstracts, not document retrieval. Documents retrieved are likely to have incomplete or irrelevant data, which generates noise.

Biomedical texts are also lengthy and complicated, with several findings within a single document. This means that relevant retrieval will not necessarily aid in providing the right response.

Advanced RAG retrieves more documents, thus enhancing noise. The CRAG architecture eliminates noise from context; however, it cannot compensate for the decline in performance once the right evidence is not retrieved.

Finally, dense retrieval processes rely more on semantic similarities than relevance to the answer, thus generating noise.

Detailed Failure Analysis However, a more detailed analysis of wrong predictions shows that such mistakes arise due to semantic drift, when the information extracted from documents is relevant to the topic but does not provide an evidence for answering the question.

Moreover, inconsistency between several documents in the biomedical literature results in ambiguity, which confuses the model.

Context dilution, when there is too much information extracted by the model, is another problem.

4.8 Visualization and Trends

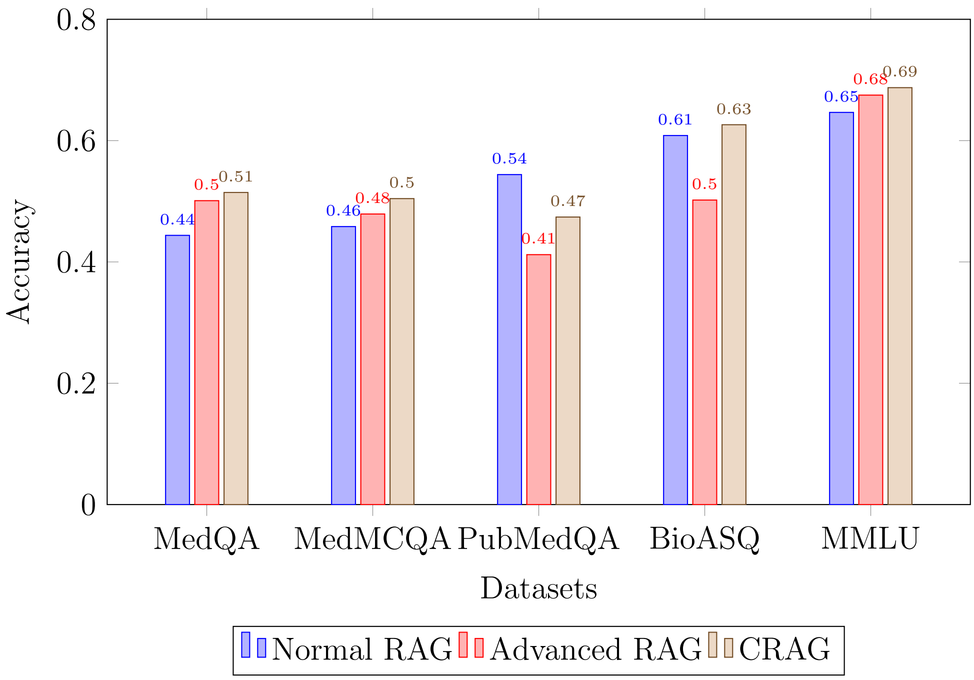

Figure 4.1: Performance Comparison of Normal RAG, Advanced RAG, and CRAG using the same model (II-Medica)

This chart shows how Normal RAG, Advanced RAG, and CRAG differ in their performances based on the same base model. Although the use of Advanced RAG boosts performance in reasoning-based benchmarks like MedQA and MMLU, it is unstable in literature-oriented datasets such as PubMedQA and BioASQ.

On the other hand, the CRAG framework is reliable in all benchmark sets since it filters unreliable contexts. This proves that enhancing reasoning with retrieval produces better performance than retrieval enhancement alone.

Trend Interpretation It is clear from the trends above that enhancing retrieval is not enough for stable performance. Although enhanced retrieval boosts performance in reasoning-based benchmarks, it destabilizes performance in literature-oriented benchmarks.

Enhancing the reasoning with retrieval in the CRAG framework resolves this issue.

4.9 Key Observations

- Retention by itself doesn't mean improvement
- Advanced RAG systems create instabilities in some datasets
- Combination of reasoning and retrieval works better than retrieval
- CRAG makes the performance more stable across various datasets, which means that it provides better performance stability as opposed to retrieval.

4.10 Discussion

The findings can be correlated with the architecture described in Chapter 3, since CRAG increases reliability through eliminating noise from context and applying reasoning capabilities.

Thus, it becomes clear that in the future, all the medical AI systems need to combine both retrieval and reasoning capabilities.

Thus, it becomes clear that besides increasing accuracy, CRAG also enhances performance consistency, which is an important factor of reliability.

To conclude, it becomes obvious that just by applying retention techniques, one cannot develop a reliable medical QA system, hence reasoning is needed.

Chapter 5

Discussion and Future Work

5.1 Introduction

The current chapter analyzes the experiment results described in Chapter 4. The aim of such analysis is the investigation of different system settings in order to reveal their advantages and disadvantages as well as important information derived from comparing various system configurations. It should also be mentioned that possible ways for further research can be found during this study.

5.2 Interpretation of Results

From the experimental findings, it is evident that the efficiency of the medical QA system is dependent not only on the model used but also on the quality of the retrieval pipeline as well as the use of the retrieved information. One major realization from the experiments is that the efficiency of the Retrieval Augmented Generator (RAG) is significantly affected depending on the dataset used.

In the case of the normal RAG, although the information extracted using retrieval is helpful in most instances, its use may not necessarily translate into enhanced results. From the experiment, there are situations where retrieval may introduce noise into the final output especially where the retrieved information is not fully related to the questions posed. Such effects are highly experienced where reasoning is key in exam-like datasets like MedQA and MedMCQA.

With the advanced RAG configuration which enhances embedding and reranking, there is an improvement in reasoning-related data sets such as MMLU and MedQA. It is interesting to note from the results that even with enhanced retrieval, the Advanced RAG performs inconsistently in literature-based QA tasks like PubMedQA and BioASQ.

The introduced framework solves these problems through an addition of the reasoning component which is responsible for assessing the retrieved context before generating an answer. Classification of retrieved context as being either agreeing, contradicting or unrelated makes it possible for the framework to exclude any misleading contexts from consideration. The result is not only higher consistency of its performance across different datasets but also its superior performance in literature-related tasks.

5.3 System Performance Analysis

Further analysis showed that there was indeed a connection between the two processes that had a strong influence on system performance. While retrieval process made it possible for the system to access external information sources, reasoning assured that only relevant and reliable data was used.

One of the most interesting findings was a clear contrast between sensitivity and robustness of systems based on either component. Purely retrieval systems appeared to be very sensitive to any changes in retrieved context. Conversely, the CRAG framework demonstrated the highest robustness among all considered approaches.

The other important finding is related to the necessity of employing hybrid retrieval. Methods based on dense retrieval take into account the meaning of terms but might not catch the exact words related to medicine, whereas methods based on BM25 guarantee that the terms are matched precisely.

The other factor that helps to achieve improvements is few-shot calibration. It is worth noting that without it, the classification will be incorrect because there is a tendency to over-classify the context as contradictory. It is called classification collapse, and providing some examples is helpful in overcoming it.

5.4 Limitations

However, there are several limitations despite the progress made through the development of the proposed system.

Firstly, the dependence on the knowledge base is still present. In cases when relevant data is not available, the ability to create any meaningful context through retrieval will be impossible regardless of the approach used.

Secondly, the classification process in the framework uses prompt-based reasoning, which can affect the performance of the model used. Even though the calibration process improves consistency, mistakes during the classification can still occur.

Thirdly, there are limitations from an algorithmic point of view. The use of large models is possible, but due to computation restrictions, it cannot be implemented in practice.

Finally, even though the system was evaluated on its ability to classify data correctly, it might make the wrong decisions for the right answers. Since a proper explanation is required in such cases, future studies should use more sophisticated assessment techniques.

5.5 Future Work

There are many ways in which the current study may be expanded.

To start with, there are other areas which might benefit from creating even larger models and applying reasoning to specific domains, such as medicine.

The process of information retrieval is one area which may be improved. It may include implementing multi-stage information retrieval methods, advanced ranking algorithms, or even indexing tailored to the specific field of medicine.

The CRAG framework itself has much potential for being expanded upon by including other types of reasoning models, such as those involving multi-agent approaches or even incorporating medical ontologies.

In addition, in order to effectively measure the performance of this system in the future, a new evaluation scheme emphasizing reasoning may be implemented.

Finally, optimization and scalability represent two other important issues related to this approach.

5.6 Practical Implications

In terms of the practical implementation of the developed algorithm, the conclusions drawn are highly applicable. False and baseless statements in relation to the sphere of healthcare may result in very grave consequences. Thus, one could safely assume that any system which does not feature validation and inference will likely lack the needed reliability to be used in practice.

However, it is clear from the analysis that by employing validation and reasoning in the course of the process, the trustworthiness of the model may be considerably enhanced. The framework may be applied to various spheres of activity including but not limited to medical advice, education, and scientific research which require accuracy on the part of the machine learning model.

Conclusions Regarding Practical Reliability Aside from increasing explainability of the process, CRAG makes it possible to develop more reliable artificial intelligence due to the availability of the reasoning stages. By analyzing the context through the lens of validation and filtering, one will gain an insight into how the machine learning model works.

Furthermore, the increase in stability is highly important as stability is a prerequisite for the effective operation of any decision-making system, especially in relation to the healthcare sector.

Chapter 6

Conclusion

6.1 Summary of Work

The purpose of this research is to examine whether the application of RAG can be useful in resolving problems within the domain of medicine, bearing in mind that it is important to have a system that is both robust and reliable. The experiment began with an initial approach, using a small-sized model, which was later replaced by more advanced models and better techniques.

Three main experimental approaches were used to test the hypothesis, namely, Normal RAG, Advanced RAG, and Agentic Corrective RAG (CRAG). As we proceeded to evolve our experiment, we could compare the effects of retrieval quality and reasoning on the system's performance.

Alongside traditional success measurements, the experiments also made use of hybrid retrieval, agentic reasoning, and trajectory evaluation methods.

6.2 Main Findings

Here is what our experiments revealed:

First of all, retrieval does not guarantee better performance. Although it helps us access additional information from external sources, it introduces noise because of partially relevant and semantically related yet not specific to answers documents retrieved.

Second, better retrieval performance is correlated with improved performance for the tasks involving reasoning, although it does not necessarily mean that the improvement will be guaranteed across all datasets. The literature-derived datasets like PubMedQA are particularly vulnerable to noisy retrieval.

Third, using reasoning via the CRAG framework substantially increases robustness. This model adds an extra layer of validation of the obtained context before generating the answer. It contributes to much better performance stability across different datasets, showing us that reasoning in combination with retrieval works much better than just improving the former.

6.3 Contributions

Unlike RAG models where the context retrieval process is considered black box, the created model involves structured categorization of retrieved information as agreement, contradiction, or irrelevance. That is how the system gets an ability to reason about the information from the outside world and does not simply apply the retrieved content as it is.

What is more, the present study provides the formalized description of the CRAG pipeline and conducts experiments for different settings of RAG in order to verify empirically that reasoning-driven retrieval improves the performance for different medical datasets.

Finally, the concept of trajectory evaluation brings a deeper understanding of the inner process and, consequently, makes the analysis easier.

6.4 Future Work

Nevertheless, the following areas should be addressed in future research despite the good performance achieved by CRAG.

The improvement of the retrieval results can be reached by applying more sophisticated ranking algorithms and domain-based search engines. It is also necessary to test multi-hopping approaches or approaches based on several iterations in the retrieval phase to improve the efficiency of the framework on complex cases.

Currently, the classification of information takes place via prompts. One may achieve better results through fine-tuning or downsizing of the model.

Extending the information base not only to PubMed Central Open Access but also to other databases providing guidelines and structured data will increase the reliability of the results.

It will also be interesting to investigate the possibility of deploying the proposed framework in real time in healthcare organizations.

Finally, it seems to be very promising to develop an adaptive model which would switch among retrieval and reasoning strategies based on the type of the task.

6.5 Conclusion

Summing up, this study proves the importance of filtering, evaluation, and reasoning with regard to information retrieved as well as just the retrieval itself concerning the improvement of medical QA systems.

That is how CRAG becomes an important advance since the framework unites the elements of retrieval and validation as well as reasoning in order to improve the reliability of AI models.

Furthermore, proceeding with this area of research, it is necessary to combine these three aspects mentioned above to get the safe AI models.

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